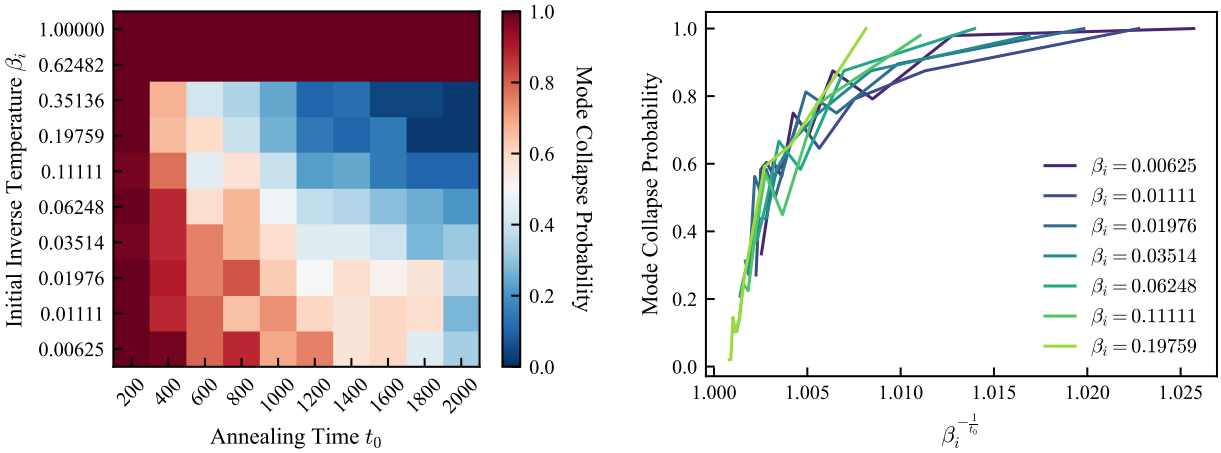


(a) (b)

Figure 1: *Annealed VI mode collapse probability with RealNVP student on a 5-mode target.* Mode collapse probabilities are computed on 50 different runs for each β_i and annealing time t_0. We employ an exponential annealing schedule (Equation (5) of the main body) on a bimodal Gaussian mixture target with hyper-paramters $d = 128$. The generative model is a RealNVP with 8 coupling layers, each containing two MLPs of depth 4 and hidden dimension 4d $= 512$. The target is a Gaussian mixture with 5 isotropic components, generated by drawing the 5 mode locations uniformly on the sphere of radius $R = 3\sqrt{2}$. One of the modes is assigned probability 0.5 while the remaining 4 are assigned probability 0.125.
fig. 1a reports the mode collapse probability as a function of (β_i, t_0), fig. 1b reports the mode collapse probability as a function of the annealing rate.

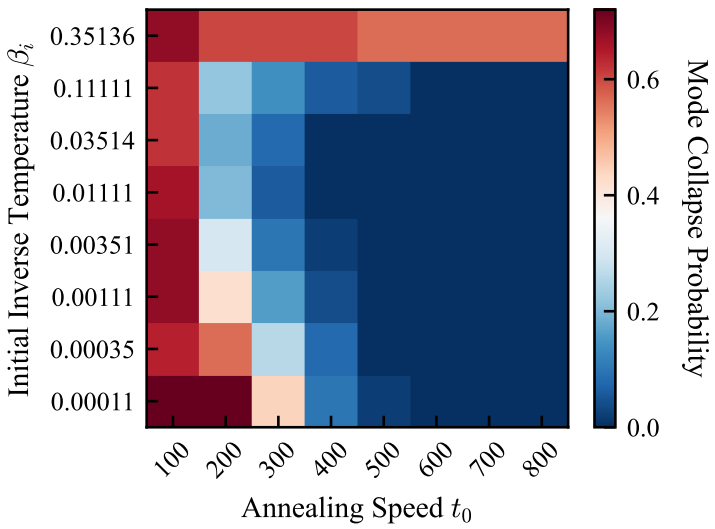
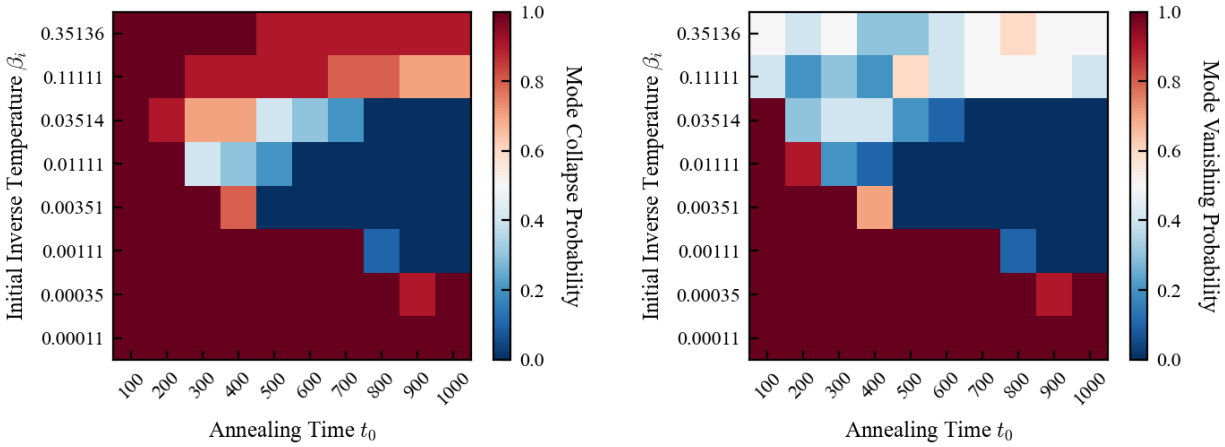


Figure 2: *Annealed VI mode collapse probability with overparametrized Gaussian mixture student.* The teacher is a bimodal gaussian mixture, the student is a Gaussian mixture with 5 components and uniform weights. The form of the annealing scheme is fixed by Equation (5) of the main body and different values of the initial temperature β_i and annealing time t_0 are explored. Hyperparamters are $d = 512$, $R = 3$ and $w_* = 0.8$. For each values of (β_i, t_0), the mode collapse probability is obtained using 50 different random student mean initialization (student means are initialized randomly on the sphere of radius R and variances are initialized at $1/\beta_i$).



(a) (b)

Figure 3: *Annealed VI mode collapse probability with Gaussian mixture student with learnable weights.* Both the student and the teacher are bimodal gaussian mixtures. The form of the annealing scheme is fixed by Equation (5) of the main body and different values of the initial temperature β_i and annealing time t_0 are explored. Hyperparamters are $d = 512$, $R = 3$ and $w_* = 0.8$. For each values of (β_i, t_0), the mode collapse probability is obtained using 50 different random student mean initialization (student means are initialized randomly on the sphere of radius R, variances are initialized at $1/\beta_i$ and weights at 0.5). We use Adam optimizer.
fig. 3a reports the probability of mode collapse (due to mean alignment or mode vanishing), fig. 3b reports the probability mode vanishing.